

Insights Document

Fake News Detection Using Advanced NLP – Internship Project

1. Purpose of This Document

This document distills the practical findings from the project's exploratory analysis and modeling comparison into a short, decision-oriented summary – the kind of artifact a stakeholder who doesn't want to read the full research report or notebooks would read first.

Note: At a glance: this project's core finding is a direct, apples-to-apples comparison of six model/feature combinations on identical train/test splits, with one dataset-specific risk (the Reuters wire-service dateline) identified and neutralized before any model saw the data. The specific winning model and its numbers populate once notebook 07 is run on the real dataset – see Section 4.

2. Dataset Insights

- The ISOT dataset is close to balanced between real and fake articles, meaning accuracy is a reasonably trustworthy top-line metric here, though precision/recall/F1 remain the primary comparison basis.
- Real articles in this dataset are overwhelmingly Reuters wire-service pieces, which almost all open with a consistent “CITY (Reuters) -” dateline. This is a dataset artifact, not a genuine fakeness signal, and was deliberately removed during preprocessing so the model can't shortcut on it.
- Article length (word/character/sentence count) shows visible separation between classes, suggesting structural/stylistic differences beyond individual word choice – worth highlighting as a finding in its own right.
- Top-word analysis by class previews the kind of vocabulary TF-IDF will lean on; least-frequent-word analysis is a useful quick check for leftover noise (typos, encoding artifacts) before modeling.
- Vocabulary size (unique words) differs between real and fake articles, which affects how large a TF-IDF vocabulary is worth keeping and how much a Word2Vec model can learn about each class's less common terms.
- Duplicate article bodies were present in the raw data and removed before any statistics were computed – an easy detail to miss that would otherwise silently let the same article appear in both train and test splits.

3. Feature Engineering Insights

Consideration	TF-IDF	Word2Vec
What it captures	Weighted word frequency	Semantic similarity between words
Interpretability	High – each dimension is a word	Low – latent dimensions
Best suited to	Linear models (SVM, Naive Bayes)	Capturing paraphrase / synonym patterns
Risk	Can overfit to rare, high-signal terms	Averaging can dilute a few important words
Dimensionality	Thousands (sparse)	~100 (dense)
Training cost	None beyond counting	Shallow neural network, still fast at this scale

Which representation wins in practice is an empirical question this project is specifically designed to answer – see Section 4 below and the full comparison in notebook 07. It's worth resisting the temptation to assume Word2Vec must win because it's the more sophisticated technique: TF-IDF is a famously strong baseline on datasets with clear vocabulary-level separation between classes, which the EDA word-frequency analysis suggests may be the case here.

4. Model Performance Insights

Note: This section is a template. Once notebook 07_Model_Evaluation.ipynb is run on the real dataset, replace the placeholders below with the actual leading model, its metrics, and the specific error pattern seen in its confusion matrix.

[Leading model]: [model name], F1 = [value], accuracy = [value].

[Key error pattern]: describe whether the leading model favors precision or recall – i.e., whether it is more prone to flag real news as fake (false positive) or let fake news through (false negative) – and what that implies for how confidently the app's predictions should be trusted at the default threshold.

[TF-IDF vs. Word2Vec]: state which representation performed better overall across the three algorithms, and whether that matches the theoretical expectation that TF-IDF tends to be a strong baseline on artifact-prone datasets like this one, or whether Word2Vec's semantic features provided a measurable edge.

[Training time vs. accuracy trade-off]: note whether the top-performing model is also the fastest to train (Naive Bayes), a moderate cost (SVM), or the slowest (Random Forest) – relevant if the model will ever need periodic retraining.

5. Practical Implications

- The system is best used as a first-pass screening signal alongside human review, not as a standalone fact-checking authority – it detects writing-style patterns, not factual accuracy.
- Because the model was trained on a specific historical snapshot of U.S.-centric English-language news, its reliability on breaking news about unfamiliar topics, non-English text, or other regions' news ecosystems is unproven and should not be assumed.
- The confidence score surfaced in the Streamlit app gives a way to triage predictions – low-confidence predictions are better candidates for manual review than high-confidence ones.
- Very short input text (the app's own threshold is 15 words) produces meaningfully less reliable predictions, since there's less linguistic signal for the model to work with – worth communicating to any end user of the app.

6. Risk Insights

- Model staleness: news vocabulary and topics shift over time (new events, new public figures, new terminology); a model trained on a fixed historical dataset will gradually drift out of step with current writing patterns if never retrained.
- Overreliance risk: presenting a confident-looking label without context invites users to treat it as authoritative; the app's design (confidence score, explanatory expander, sidebar disclaimer) is a direct mitigation, not a formality.
- Adversarial adaptation: if this kind of classifier were used at scale, producers of fake content could adapt their writing style specifically to evade detection – a known dynamic in the fake-news-detection literature and a reason not to treat any single model as a permanent solution.

7. Recommendations

- Adopt the F1-leading model/feature combination identified in Phase 11 for the deployed app.

- Periodically re-evaluate the model against newer articles to catch performance drift as news topics and writing styles evolve.
- Consider the documented stretch improvements (TF-IDF-weighted Word2Vec averaging, transformer-based models) if this project is extended beyond its current internship scope.
- If this project is ever extended toward real-world use, validate against a second, independently-sourced dataset before trusting performance numbers from ISOT alone, given the source-concentration risk noted in Section 2.